Waterwatch Tasmania

Reference Manual

Part 3

Developing a Monitoring Plan

Developing a Monitoring Plan

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Introduction

The *Part 3 Reference Manual: Developing a Monitoring Plan* is designed to help the community networks to ask the right questions and collect the right data in the time and resources available.

Based on the outcomes of your catchment survey from *Part 2 Getting to Know Your Waterway and Catchment*, you will by now have decided that monitoring is needed to answer important questions about the health of the waterway. You will also know what resources are available and the interests and capacities of group members. You are ready to plan your monitoring.

A monitoring plan is both a guide to, and record of, decisions about the design of your study. A good monitoring plan will ensure that you ask the right questions and collect the right data in the time and with the resources that you have available. Your plan will help you choose the best monitoring tools for the task and define the scale and complexity of your monitoring work. A well thought out plan at this stage could save an enormous amount of wasted effort later on.

A monitoring plan is built around eleven questions. These are:

1. Why are you monitoring?
2. Who will use your data?
3. How will the data be used?
4. What will you monitor?
5. What data quality do you want?
6. What methods will you use?
7. Where will you monitor?
8. When and how often will you monitor?
9. Who will be involved and how?
10. How will the data be managed and reported?
11. How will you ensure that your data are credible?

All groups will find value in answering the eleven questions, but the depth and extent of your answers will depend on the purpose of monitoring, the skills of your group and your resources. The answers to these questions are not strictly sequential so you may need to revisit earlier questions as your ideas develop: take some time to consider each question carefully.

An important question to consider when planning to monitor is who will use the data. In general, if the data is to be used primarily for education or raising awareness of group members, then the quality of the data is secondary to the process of collecting it. However, if data is to be collected for decision making by the group or is to be forwarded to an external organisation such as local government, then you should plan to collect data with a demonstrable level of quality.

There are two important considerations that influence the level of confidence that others will have in your data - data confidence measures and quality control checks.

- *Data confidence* measures refer to those broad activities for maintaining quality in all aspects of monitoring, e.g. training, keeping written records, quality control. It will ensure that results from the many hours of monitoring work will lead to achieving the goals set by your group and a greater sense of personal satisfaction. It will also allow you to demonstrate the quality of your program to others so that both you and they can have confidence in the data.

- *Quality controls* (QC) are those activities you do to ensure that the amount of error in your results (how far from the real result you are) is acceptable for the question you want to investigate. You will be able to measure the quality of your data and show others just how good your data are.

This guide includes the following sections:

- Section 1 The short monitoring plan.
  The eleven questions in the **Short monitoring plan** provide a useful checklist to plan an effective visit to the waterway. The plan is suitable for groups starting to plan for the first time and will help to ensure effective education and awareness raising activities.

- Section 2 The extended monitoring plan.
  If you are aiming to collect data that will help answer important questions about the condition of the waterway, this section is for you. Planning in depth will help you to arrive at findings and conclusions in which you and others can have greater confidence. Your Extended monitoring plan will include data confidence measures and will both guide and record your decisions during this important stage.

Section 1. The short monitoring plan

This eleven question plan provides a solid foundation should you wish to develop further (See **Section 2. The extended monitoring plan**).

1 Why are you monitoring/surveying?

The first step in planning is to ask why do we want to monitor? Answers may vary but often groups simply want to know what the stream is like.

2 Who will use your information?

Potential users might include students and/or group members, etc. Name the main groups.

3 How will the information be used?

Data could be used for more than one purpose, e.g. to educate students about the principles of ecology or identify major trouble spots in the waterway. Knowing its main use will help determine the right kind of data to collect. Describe its intended use.

4 What will you monitor?

The things you choose to monitor will depend upon the question(s) you are asking as well as the resources available. For example, if your group wants to learn about the general ecological health of the waterway, then the main types of water bugs (macro-invertebrates) present could tell an interesting story. List the things that you will monitor. See **Table 1. General guidelines for monitoring** for a full list of things that can be monitored.

5 How accurate should the data be?

This depends upon the question(s) you are asking and how you intend to use the data. At the very least your data should be accurate enough to indicate the location of grossly polluted sites. Depending on your findings you may then choose to refine your monitoring program (see **Section 2. The extended monitoring plan**). For groups with a focus on education and raising awareness, the quality of the data is secondary to the actual process of collecting it.

6 What methods should be used?

This depends on your objective(s) and resources. There are often several ways of testing the same parameter, e.g. for high precision turbidity readings from 0 to 1000 NTU, a turbidity meter costing several thousand dollars is required, but a turbidity tube (≈ $35) is suitable for less precise readings between 10 - 400 NTU. Use the same methods at all sites to allow comparison of data. List which methods will be used.

7 Where will you monitor?

The location of monitoring sites depends on whether you are monitoring a river, lake, estuary or ground-water, and also on the purpose of monitoring. For example, monitoring at a variety of typical sites in the catchment is good for providing information about its overall condition. On the other hand, sites located above and below a pollution source are needed to indicate its impact. Your sites should be representative of the condition of the waterway. Use a map to show your sites.

8 When will you monitor?

This depends on your resources and the purpose of monitoring. For example, if you are interested in a snapshot of the waterway, then monitor a number of sites on the same day. Monitoring pollution events, e.g. discharges from pipes, depends on the timing of the discharge and surveying the physical form of the stream is best done during low flows for safety reasons. Describe when you will monitor.

9 Who is going to be involved and how?

Indicate who will carry out surveys and/or test water samples, who will arrange transport to sites and back, who will prepare the water testing equipment to be used, who will photograph sites, etc.

10 How will monitoring data be managed and presented?

It is important to record and present the data. It helps to raise awareness of the condition of the waterway amongst members and helps you to refine your monitoring activities. Identify who will look after the data and describe how it will be managed.

11 How will you ensure that your information is credible?

Developing answers to the first ten questions above is the first step in conducting an effective visit to the waterway. For all surveys and tests, ensure that group members are adequately trained. For water quality tests, ensure that any instruments used are calibrated and read correctly and that any water samples from rivers are taken from the main current at about 25cm below the surface. List what you will do to improve the credibility of your information.

As you worked through the questions above, you may have found that a more thorough plan is needed to effectively gather data to answer the concerns of your group. In this case, go to **Section 2. The extended monitoring plan**.

Table 1. General Guidelines for monitoring

How to use this table

1. Identify on your catchment map the potential pollution sources and/or different land uses from the headwaters to the mouth.
2. For each potential pollution source and/or land use read along the row in the table to determine the best things to test, where, when and how often.
3. Choose tests and surveys as determined by your group's expertise, equipment and objectives.
4. Record your decisions in your monitoring plan.

KEY

Conductivity (salinity)	NO_3 - nitrate	Metal - heavy metals
DO - dissolved oxygen	$PO_{4\,(o)}$ - orthophosphate	Temp - temperature
FC - faecal coliform bacteria	$PO_{4\,(t)}$ - total phosphate	

Main purpose	Comments	Critical things to test or survey				Locating your sites			When do you test?	
		Physical/chemical tests	Macro-invertebrates	Algae	Habitat	Reference, impact and recovery sites	Sample in paired catchments	Other areas to test	Regular intervals	Other times
Establish a baseline	Essential if condition of waterway is not known.	pH Turbidity Temp Flow Conductivity Velocity $PO_{4\,(o)}$ NO_3 DO flow $PO_{4\,(t)}$ FC	✓	✓	✓			Choose representative sites, bottom of main tributaries.	Weekly - monthly for physical / chemical tests during base flow conditions. Twice yearly for macro-invertebrates.	
Determine suitability for particular uses:										
Protection of aquatic ecosystems	Ecosystems are affected by many pollutants and clearing of riparian vegetation.	Turbidity Temp pH Conductivity $PO_{4\,(o)}$ NO_3 DO PO_4 (t)	✓	✓	✓			Sample within water body to be protected.	As above	
Drinking water	Quality determined by chemicals, bacteria and taste.	Turbidity Conductivity pH taste NO_3 FC		✓	✓			Sample drinking water.	Weekly to monthly for physl/chem tests.	
Recreation	Bacteria and aesthetics are main problems.	Turbidity FC		✓	✓			Sample at recreation site.		During times of recreational use.
Agriculture	Farm productivity may be affected by poor water quality.	Conductivity pH NO_3 Pesticides FC		✓	✓			Sample water used for agriculture.	Weekly to monthly for physical/chemical tests.	Sample water used for irrigation during summer.

Main purpose	Comments	Critical things to test or survey						Locating your sites			When do you test?	
		Physical/chemical tests			Macro-invertebrates	Algae	Habitat	Reference, impact and recovery sites	Sample in paired catchments	Other areas to test	Regular intervals	As necessary
Assess impact of land uses / pollution sources:												
Forest practices	Roading, clearing and fires can lead to soil erosion and algal growth. Herbicides may be used.	Turbidity Velocity	PO_4 (o) NO_3 Flow	Pesticides	✓	✓	✓	✓	✓			During rain
Urbanisation / stormwater	Runoff contamination and flooding are common problems.	pH Turbidity Velocity	PO_4 (o) NO_3 DO, Flow	FC, metals pesticides	✓	✓	✓	✓		Sample at run-off points.	✓	During rain and discharge events.
Livestock operations	Manure, bacteria and nutrients from feedlots impact waterways.	Turbidity	PO_4 (o) NO_3 DO	FC	✓	✓	✓	✓			✓	Per discharge from feedlot and rain.
Cropland / pastures	Soil erosion from heavy grazing. Fertiliser or herbicide runoff and salinity problems.	Turbidity Conductivity Velocity	PO_4 (o) NO_3 Flow	PO_4(t) FC	✓	✓	✓	✓	✓	Sample within cropping areas. Sample ground water.	✓	During rain and after fertilising.
Mining operations	Sediment, tailings, dust, chemicals can have very long term impact.	Turbidity Conductivity pH, velocity	DO Flow	Metals	✓		✓	✓		Sample at single point discharge sites.		During rain and discharge events.
Construction sites	High sediment and chemical runoff from poorly managed sites.	Turbidity Conductivity pH, velocity	Flow	Metals	✓		✓	✓		Sample at run-off points.		During rain and discharge events.
Septic systems	Leaks, overflows and leachate can have severe effect on quality and cause health problems.		PO_4 (o) NO_3	FC	✓	✓		✓		For lakes, sample near and away from pollution source.	✓	During times of high demand, rain and recreational use of waterway
Golf courses and playing fields	Runoff carries nutrients and pesticides.		PO_4 (o) NO_3	Pesticides	✓	✓	✓	✓	✓	Sample at runoff points.	✓	During rain
Dams	Changes in flow rates during filling or releases stresses aquatic ecosystem. Low DO release water is a problem.	Turbidity Temp pH Velocity	DO Flow		✓		✓	✓		Profile temp and DO from top to bottom.	✓	During filling of dam or release of stored water.

Section 2. The extended monitoring plan

In Section 2, the eleven questions have been broken up into their component parts to help coordinators facilitate planning by community groups.

Answering these questions is not easy. Start by establishing a planning committee composed of your coordinator, data users, key group members and scientific advisers. Through the committee, you can plan to collect useful and relevant information.

It is helpful to carry out a short period of monitoring as a reconnaissance to reveal the character of your waterway. The reconnaissance study will help you to define the key issue, problem or question to be answered by monitoring and assist with developing the monitoring plan.

Answers to the questions are likely to change as your program matures. For example, your group might find that a method is not producing data of high enough quality, data collection is too labour-intensive or expensive, or additional indicators need to be monitored.

See an example of an extended monitoring plan in **Appendix 2. Example monitoring plan**.

1. Why are you monitoring?

There are many questions to consider such as what to sample, when and where to sample, how to test the samples, etc. These cannot be answered without knowing the main reasons for monitoring. Your reasons can take the form of objectives or questions to be answered. Your reasons for monitoring will guide your answers to all other choices so it is worth considering these carefully. By choosing your question or objective carefully you will avoid difficulties such as:

- solving the wrong problem;
- collecting irrelevant data;
- asking a question in such a way that no solution is possible;
- accepting the wrong solution before understanding the problem.

Table 2. Purpose of monitoring and indicators of condition of the waterway gives a general outline of common reasons for monitoring and the general indicators to measure. They provide a starting point for identifying the monitoring purpose for your group. The reasons for monitoring will determine the level of resources and training required by your group.

Table 2. Purpose of monitoring and indicators of condition of the waterway

General reasons for monitoring	Monitoring approach
• Identify degraded sites. • Determine impact of land uses. • Identify water uses, users, diversions and stream obstructions, and barriers to fish movement. • Snapshot of physical and riparian condition to benchmark future changes.	Survey of catchment for riparian vegetation, physical form of stream. See *Part 2 Getting to Know Your Waterway and Catchment.*
• Screen for pollution. • Identify severity of pollution impacts from land uses, e.g. forestry. • Identify effectiveness of pollution controls and river restoration activities, e.g. planting riparian vegetation. • Identify water quality trends.	Macro-invertebrate sampling and algal sampling. See *Part 5 Monitoring Algae* and *Part 6 Macro-invertebrate (Waterbug) Monitoring.*
• Screen for pollution and identify pollution sources. • Identify specific pollutants of concern, e.g. excess nutrients leading to algal blooms. • Identify water quality changes downstream, e.g. salinity leading to unacceptable water for stock or crops. • Determine whether water quality meets minimum standards for human uses, e.g. recreation. • Determine whether stream quality meets values and uses desired by community (protected environmental values), e.g. drinking water.	Water quality sampling See *Part 4 Water Quality Testing.*
• Show government officials that the local community cares about the condition and management of its water resources. • Educate the local community and encourage stewardship of the environment. • Provide students with skills, experience and understanding of the methods of environmental science. • Determine the state of the rivers, e.g. State of Environment reporting.	All of the above.

Defining the problem or key question to be answered is a process that should involve the users of your data. See **2 Who will use your data?**

As you consider the remaining questions in this section you will progressively fine tune the problem or question for monitoring. In the end, the monitoring objective should:

- be specific;
- say what is to be measured;
- be achievable given the skills of the group and resources available;
- indicate when monitoring will be completed;
- be meaningful.

Poorly written monitoring objectives are vague, obscure, not measurable, unrealistic, trivial or about activity rather than results.

A regional community group may identify several complementary objectives for monitoring which match the individual needs of schools, individuals, local government and others. What is important is that group members monitor for a number of very valid reasons. Groups must clarify the reason for monitoring so they can take the necessary steps for success.

Note that professionally designed programs often state the monitoring objective as a testable hypothesis which can be either rejected or accepted based on statistical analysis and 95% confidence levels. Since this is a complex field, community groups generally do not attempt to test hypotheses but aim for a more descriptive study of the waterway.

1.1 List the specific objective/question(s) you will try to answer through monitoring.

1.2 Add brief relevant background information to your monitoring plan.

Use the information in your catchment study as a source for a summary of background information.

2. Who will use your data?

Your group should consider all potential data users when planning. As government agencies recognise the valuable contributions of community groups and the necessity of doing more work with fewer resources, there is a shift taking place in the way environmental data is collected. Scientists and decision makers are recognising the value of community participation in all aspects of environmental policy and planning.

Knowing who will use your data is helpful in developing a successful community network. Some potential data users are listed below:

- **Users with widely varying data requirements:**
  - group members;
  - teachers and students;
  - environmental organisations.
- **Users of medium to high quality numerical and/or qualitative data:**
  - local planning officers in councils;
  - government departments;
  - parks and wildlife staff.
- **Users with high quality numerical data requirements:**
  - state or local government water quality analysts;
  - universities and other educational institutions.

As the group monitoring project is being planned, your coordinator should contact as many potential information users as possible to determine their data needs. Find out from data users under what circumstances they will use your information and invite their participation in planning to monitor. It is important to have at least one user committed to receiving and using the data. In many cases that user will be the group itself.

2. List the main users of your data.

3. How will the data be used?

Community monitoring data can be used at different scales.

- At the individual level, data builds awareness and contributes to a sense of ownership of the waterway.
- At a local level, data can be used to influence developers, councils and community members.
- At the catchment level, information can be used to link upstream and downstream stakeholders and identify degraded waterways.

The range of uses of data is limited only by the imagination.

- For many groups, the value of monitoring work is not so much in the data it produces, but rather in promoting community involvement, building awareness and a sense of stewardship towards the environment, and educating young people. Volunteers gain a better understanding of their environment by going down the river.
- For other groups, the data being collected are meant to be more of a "qualitative" indicator – highlighting hot spots and red flags for follow-up by agency staff. Data at this level can be used for setting priorities.
- At a more technical level, data which meets the *Part 7 Data Confidence Guidelines* can be used to influence local planning decisions, e.g. where to site a sewage treatment facility.

Your data may also be used for:

- developing a catchment management plan;
- measuring the effectiveness of river restoration work;
- State of the Environment reporting;
- screening waters for potential problems;
- gathering baseline information about the state of the river;
- answering specific research questions;
- assessing non-point pollution levels.

You should find out what will make your data suitable for its intended use. This will help determine the right kind of data to collect and the level of effort required to collect, analyse and report it. Planning may require a high data confidence level i.e. the data should be reliable.

3. Describe how the data will be used.

4. What will you monitor?

A waterway is a very complicated system of inter-related physical, chemical and biological characteristics. A lack of coordination between groups providing the data and those using the data, and unclear identification of what is needed and why, can lead to the wrong parameters being monitored. Commonly monitored parameters, their reasons for monitoring and the methods used to measure them are shown below in **Table 3. Commonly used parameters and what they can tell us**.

Table 3. Commonly used parameters and what they can tell us

Parameter	Reason for monitoring	What methods to use
Stream habitat condition	Habitat quality affects health of the aquatic ecosystem and human uses downstream.	See **Habitat Rating** in *Part 2 Getting to Know Your Waterway and Catchment.*
Macro-invertebrates	Diversity of aquatic macro-invertebrates help to indicate health of waterway.	Use nets or artificial substrates to collect aquatic macro-invertebrates and assess on site or back in the lab. See *Part 6 Macro-invertebrate Monitoring.*
Algae	Algae provide a general indication of health, also over-enrichment of stream with nutrients. Presence of blue-green algae indicates risk to health and poor condition of the catchment.	Field survey of algae and a range of simple tests to determine if they are blue-green algae. Carry out cell counts in a lab. See *Part 5 Monitoring Algae*.
Faecal bacteria	Presence of faecal bacteria may indicate the presence of other disease-causing bacteria, viruses and parasites.	Collect sample from waterway and test in the lab using membrane filtration or direct incubation of samples in selective media. See *Part 4 Water Quality Testing*.
Conductivity (salinity)	Dissolved salts in the water (conductivity) affect the survival of aquatic life.	Use a conductivity meter to record electrical conductivity levels on site. See ibid.
Dissolved Oxygen	Oxygen in the water is essential for the survival of most organisms. It is an indicator of organic pollution and over enrichment of lakes.	Use an oxygen meter or chemical kit, e.g. modified Winkler method kit on site. See ibid.
Biochemical Oxygen Demand	BOD indicates the amount of oxygen consumed by decomposition of organic wastes, e.g. sewage. It measures the load of organic pollution in the river.	Use an oxygen meter or chemical kit, e.g. modified Winkler method kit on site. See ibid.
pH	Acidity of the water affects the survival of aquatic life. Indicates pollution and acidification.	Use a pH meter or pH test paper on site. See ibid.
Phosphates	Amount of phosphate in the water indicates the nutrient status, organic enrichment and consequent health of the water body.	Use a phosphate colour comparator or colorimeter on site or in lab. Very low levels need to be tested by professional labs. See ibid.
Nitrates	Amount of nitrate in the water indicates organic enrichment and consequent enrichment of the water body.	Use a nitrate colour comparator or colorimeter on site or in lab. Very low levels need to be tested by professional labs. See ibid.
Temperature	Rapid temperature changes of the water stress aquatic life and affect dissolved oxygen levels.	Use a temperature probe or thermometer on site. See Water Quality Testing.
Turbidity	Cloudiness of the water caused by suspended particles affects the survival of aquatic life. It indicates erosion and habitat destruction.	Use a turbidity tube, Secchi disk, colorimeter or turbidimeter on site. See ibid.
Discharge (flow)	Discharge (flow) and velocity of water affect pollution loads and the survival of aquatic life.	Time floating object travelling over a given distance, measure stream-bed cross-section, calculate discharge or use gauging station. See ibid.

DEVELOPING A SHARED UNDERSTANDING OF THE AQUATIC ENVIRONMENT

An important step in developing your monitoring plan is to develop a shared understanding of the main components of interest in the aquatic environment and how they interact. This understanding can be written down as a model.

Models help to show:

- the main components and linkages;
- the main problem or question to be answered by monitoring;
- cause and effect relationships;
- what to measure.

A model is a simplification of the real world and can be illustrated as a set of components (boxes) and linkages (arrows). The model shows those factors that are thought to be causing changes in the system and the consequences of changes. A box can represent a stock or quantity and an arrow can represent a flow between stocks or a process that links two boxes. For an example see **Figure 1. Model of a river showing the impact of nutrients on water quality and the aquatic ecosystem**.

Group members, scientists and others with different backgrounds, will often have different opinions about what is important in the aquatic environment. Discussing these differences is a powerful tool in developing a shared understanding of the environment. Your model need only include the components of the environment that are important for the question being investigated. If discussion leading to a shared model is not undertaken, then disagreements will occur about what to test and what data are important. Once a model has been agreed to, many of the questions about monitoring become clearer.

Models are simplifications and may be imperfect, but are a useful basis for planning. If your data seem inconsistent with the model, it may lead to better understanding of factors affecting the waterway and an improved model.

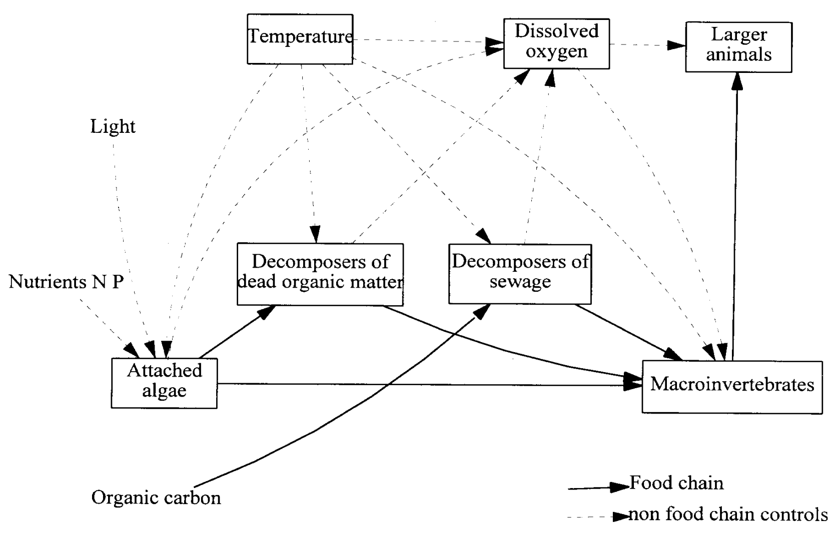


Figure 1. Model of a river showing the impact of nutrients on water quality and the aquatic ecosystem.

4.1 Describe your model using boxes and arrows.

CHOOSING THE PARAMETERS TO MONITOR

Determining what to monitor will depend upon the question(s) you are asking, your agreed model, the needs of the data users, how the data will be used and the resources of your group.

The type of question you are asking will determine the range of suitable parameters to choose from. For example, asking about the impact of water pollution on public health, could lead your group to recording the incidence of water related illness in the community. Monitoring human health indicators is not easy but can be a valuable complement to other aspects of catchment monitoring. This type of monitoring has the potential to create public interest in your monitoring work because the results are usually very relevant to people living in the catchment.

Many monitoring studies include testing of parameters that are not directly relevant to the model and which do not help to answer the question or objective. Below are some things to think about when choosing indicators/parameters.

- Does the parameter relate to the model of the waterway?
- Does it provide early warning of change?
- How difficult is it to measure with an acceptable magnitude of error?
- Do you have the resources to measure the parameter?
- Is it explainable to the users of your data?
- Are there existing records held by local or state government agencies?

4.2. List the parameters that you will survey or monitor. Describe how they will help you answer your question(s). Indicate if any are critical to the project or are collected for background information.

5. How accurate should the data be?

Deciding what data quality is needed to meet your monitoring goals is not easy! It depends on a lot of things such as why you are monitoring, the intended use of your data, the advice of data users and how skilled your group has become. The best way to get at this answer is to talk to people about why you are monitoring and the intended use of the data.

Government agencies spend considerable resources on obtaining very accurate data for making decisions, e.g. to determine if an effluent is violating discharge permits. It is much harder to collect more accurate data than less accurate data.

TOLERABLE ERROR RANGE

Error is defined as "how far our measurement could be from the truth for that specific sample, either as percentage of the value (e.g., plus or minus ten percent, or ± 10%) or as an increment of the value (e.g., ± 0.3 pH units), depending on the method." (Katznelson R, 1998).

However, you do not always need a scientific instrument that displays more digits and decimal places to make the measurements better. **The key question is - what range of error can be tolerated and still satisfy your monitoring goals?**

There are many times when indicative data are more useful than very accurate data. For example, ten inexpensive field tests with an error of ±25% will yield far more useful information about the health of the catchment than two costly laboratory analyses with a 5% error.

The size of the tolerable error depends on the purpose of monitoring and the importance of the difference or change in a parameter to be measured. For example, a tolerable error range for dissolved oxygen monitoring of ± 0.4 mg/L in the range of 5-7 mg/L is recommended if the monitoring goal is to determine whether there is sufficient oxygen in a creek to support fish life. Test equipment with a wider error of ± 1 mg/L is unsatisfactory as it may not distinguish between two samples of 4 and 6mg/L DO. This difference however, may mean life or death for the fish.

The monitoring goal chosen by the group determines the acceptable tolerable error range which in turn determines what testing equipment is appropriate. If groups cannot test within the chosen tolerable error range then either the monitoring goal or the equipment needs to be changed. See the *Part 7 Data Confidence Guidelines* for detailed help on choosing the tolerable error range.

If you monitor to identify pollution hotspots, to understand the processes going on in the waterway, or to determine its overall condition, you can use relatively simple inexpensive methods with a relatively wide error range but which still yield data that meets the needs of your group, agencies and others.

EQUIPMENT AND METHOD

Resolution is the smallest change in a parameter that your equipment method will discern with confidence. Your equipment should be able to resolve changes smaller than the designated tolerable error range. **Detection limit** is the lowest concentration that your method will report as positive. **Sensitivity** is a term often used to describe both resolution and detection limit and should therefore be avoided. You may need to trade off desired error tolerance range and resolution against the cost of methods and equipment, and skills of group members.

5.1 Describe the reporting units, resolution and detection limit and required tolerable error range of the method used to test each parameter.

COLLECTING REPRESENTATIVE AND COMPARABLE DATA

The natural variability of water bodies, e.g. from edge waters to the main current, creates a challenge for the sampler who is aiming to collect meaningful data. The data are more meaningful (better data quality) if sampling is **representative** and **comparable**.

- *Representativeness* is the extent to which your data actually reflect the conditions in the river. Unrepresentative samples are a major source of error that can far exceed the resolution of the test equipment. For example, dissolved oxygen (DO) is critical for fish survival in rivers. Monitoring DO during the most critical times, e.g. at dawn in the remaining stagnant pools at the end of summer, will detect the worst case conditions which might otherwise be missed with routine monitoring, e.g. 9am each Monday over summer. Monitoring at critical times will ensure the data are representative.

- *Comparability* is the degree to which different methods, data sets and /or decisions agree or are similar. Sample results are comparable if they have been collected under the same conditions and analysed in the same way. For example, macro-invertebrate results from one riffle should be compared with those from another riffle site but not with those from an edgewater. This is because the macro-invertebrate community found in edgewater habitats naturally differs from that found in riffles.

At the very minimum, all community groups should aim to collect data that is representative of conditions in the waterway by choosing sampling sites carefully and taking samples according to guidelines. Your desired data quality goals may need to be modified according to the equipment and resources available to your group.

5.2 Describe how your sampling will give representative and comparable data.

6. What methods should be used?

The methods adopted by your group depend primarily on your choice of indicators and parameters and the data quality required. Methods include:

- surveys - field observations of riparian vegetation and physical condition of waterways;
- macro-invertebrate sampling and assessment on site or in a lab;
- algal sampling and assessment in a lab;
- water quality sampling and analysis - samples are collected and tested in the field or in the lab.

SCIENTIFIC AND PRACTICAL CONSIDERATIONS

There are a number of scientific and practical considerations influencing your choice.

Scientific considerations

- Will the method produce data of the right quality?
- What error range applies to data produced by the method?
- What is the resolution?
- Will it measure the parameter in the range you need?
- Will the method give representative results?
- Is it comparable with methods used by government agencies?

Practical considerations

- Do you have the resources to do it?
- How difficult is it?
- How time consuming is it?
- How expensive is it?
- Will it produce data that satisfy your monitoring goals?

HABITAT SURVEYS

Field surveys of habitat quality are designed to look for long term changes in the physical condition of the waterway. Check the Reference Manual Getting to Know Your Waterway and Catchment for more information.

WATER QUALITY SAMPLING AND ANALYSIS

Macro-invertebrate and algae sampling will yield information about the impacts of point source or diffuse pollution on the aquatic ecosystem. For more information, see *Part 6 Macro-invertebrate (Waterbug) Monitoring* and *Part 5 Monitoring Algae*.

Sophisticated physical and chemical testing methods usually yield more accurate and precise data (if properly carried out), but they are also more costly and time consuming. See the *Waterwatch* Equipment Guide for a list of equipment that will meet your needs, the Test instructions for a detailed description of procedures, and the Reference Manual: Water Quality Testing for more information.

6.1 Describe your water quality sampling and analytical methods.

For each parameter that you have chosen, briefly describe the six attributes listed below.

- Parameter - what you are measuring.
- Collecting procedure - how it is being sampled.
- Container - what your water quality sample goes into. This will not apply to macro-invertebrate sampling or habitat assessments.
- Preservation method - how you keep the water quality sample from changing if transporting it to a lab.
- Maximum holding time - how long the water quality sample will last. See notes on each test in *Part 4 Water Quality Testing.*
- Equipment and method - what equipment and method is used to measure the parameter.

KEEPING TRACK OF SAMPLES

Some samples may be transported to a laboratory for further analysis. Use labels that avoid confusion and keep track of the samples from beginning to end. Every sample that is collected in the field should be labelled to record the:

- site code - a unique site code identifies where the sample was taken;
- sample number - a unique number for the sample;
- date and time of collection - hour / day / month / year;
- sample type - what will be analysed;
- sampler's name - leader or group members;
- preservation method - if any.

7. Where will you monitor?

Choosing geographic boundaries that delimit your monitoring work is important since selecting the wrong boundaries will focus your efforts away from important causes or effects. Catchment boundaries are useful for investigating the human impacts on rivers, although human activities may impact on groundwater supplies that extend over a wider area.

To get useful information it is very important to choose your sites well. The choice of sites will depend on whether you are sampling lakes, rivers, estuaries, or groundwater, what you want to know and what parameters you're measuring. For example, if you want to determine the overall character of the waterway, sampling stations should be located at a variety of sites that represent the variety of conditions in the catchment. On the other hand, if you want to measure the impact of discharge from a drain, sites should be chosen upstream and downstream to isolate the impact of the drain. Detailed guidelines for selecting sites in rivers, lakes, estuaries and groundwater appear a little later in this section.

SITE CODES

A site code uniquely identifies your site in your basin or state, e.g. TAM 010. It is a three letter - three number code. The first three letters of the code refers to the name of the water body, in this case the Tamar River, and the three numbers refer to the location of the site within the catchment. As a rule, numbers chosen in the site code should increase for sites further down the catchment. You will need to use the code when entering data into the *Waterwatch* database program. Your State *Waterwatch* Coordinator will provide your group with site codes that are unique within the State. This will allow future merging of databases in the State.

If you intend to use a site repeatedly, it will be helpful to mark the site in the field, so that you and other group members can easily find it again. One way is to attach a plastic sign to a wooden stake. Mark the sign with a waterproof pen, e.g. Group monitoring site ‘X’, site code, phone number and the group’s name.

SELECTING SITES, GENERAL GUIDELINES

1. Use your catchment map to select possible sites that appear to meet your needs.
2. Get permission to enter private land.
3. Go to each site and check that it is accessible and safe to work at, that it is representative of conditions in the waterway and suitable to your needs.
4. Photograph each site at the sample collection point and record directions to the site for future visits. Place the directions and photographs in a looseleaf binder and store your information with your monitoring plan and catchment map.
5. Add site locations to your catchment map.
6. Record details of the site location in your monitoring plan.
7. List all the sites selected, along with your reasons for choosing them.
8. Describe the precise location of the sampling point for others to do repeat sampling, e.g. “at overhanging rock on left bank 10m upstream of bridge”.
9. Register your site with your group coordinator who will allocate a unique site code.

Guidelines for selecting sampling locations for each type of water body are discussed below.

RIVERS AND STREAMS: SELECTING SITES

General guidelines

The flow of water downstream (and pollution) will influence your choice of sites. Some general considerations are listed below.

- Sampling stations should be located at a variety of sites that represent the variety of conditions in the catchment. These might include:
  - waters located in areas of different land uses, e.g. housing, agriculture, forests;
  - streams and rivers located in the upper, middle and lower catchment;
  - waters receiving point source pollution discharges and diffuse pollution.
- Where possible, choose sites that are monitored by government agencies.
- Sites at habitats of rare, vulnerable or endangered species may be sampled but consult your local nature conservation experts for advice.
- Consider choosing sites at areas of public recreation, e.g. swimming areas.
- Sites should be safely accessible. Avoid steep, slippery, or eroding banks or sites where you cannot get landowner permission to enter.
- Because conditions in the main stream may differ from those at the edge, sites should be located in the main current and away from the banks. If that is not possible, locate the site next to the bank where thorough mixing of the water occurs, such as on an outside bend of the river.
- Be aware that variable flows caused by dams, weirs and similar structures may make your site unrepresentative of water quality in the stream.

Sites for measuring point source pollution impact

Point source impact sites include stormwater drains, sewer overflow points, factories, construction sites, livestock feedlot runoff points, septic tank systems, eroding banks, and polluted tributaries to the main stream. Generally, three sites should be chosen to "bracket" the point source, see **Figure 2. Three sites for measuring point source pollution impact**:

- *a reference or control site* immediately upstream of the potential pollution source;
- *an impact site* immediately downstream of the area at the point where the pollutant is completely mixed with the water, e.g. 100m downstream;
- *a recovery site* downstream, e.g. 2km, to see how far the pollution impact extends.

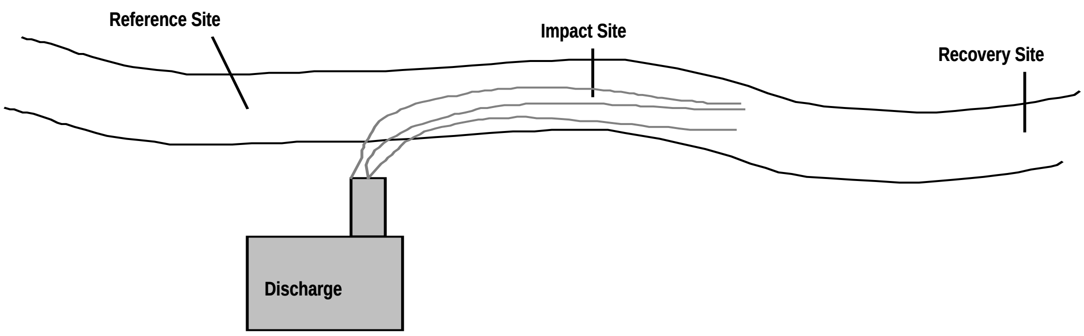


Figure 2. Three sites for measuring point source pollution impact.

Each site should be similar in all important respects to the impact site except that they are not subject to the disturbance. This ensures that differences between the control and impact sites are due to the pollution itself and not natural variation.

Sites for baseline monitoring

Sites should represent the typical conditions in the catchment. Divide the river into areas of similar condition (reach), e.g. the river may wind through steep hills and bush for 10km before flattening out into farmland. Monitor at sites representative of each reach.

Sites to determine suitability for a particular use

This depends on the use. For example, for swimming areas, sample water from the swimming hole.

Sites for macro-invertebrates

Sites for macro-invertebrate collection should be shallow "riffle" areas less than knee deep with current between 0.1 and 0.5 metres per second, and rocky/gravelly bottoms. You can also monitor edgewaters and pools but riffles have the greatest variety of habitats for macro-invertebrates and are preferred.

It is very important that all the sites be as similar as possible in every way except for the impact you are measuring: e.g. for comparability, all three sites should be riffles not a mixture of riffles and pools.

Sites for whole catchment impact

This involves pairing up an impacted catchment with a relatively natural catchment *(paired catchments).* See **Figure 3. Location of sample sites for a paired catchment study.**

For example, a small sub-catchment may be entirely converted to forest harvesting but a nearby sub-catchment is untouched. The catchments should be similar in every way except the impact being measured.

At least two sites should be chosen to compare the water quality of the impacted with an unimpacted sub-catchment:

- *a reference or control site* on the unimpacted tributary immediately upstream of where it joins the main flow (confluence);
- *an impact site* in the impacted tributary immediately upstream of the confluence.

Two further sites can be chosen to measure the impact of the sub-catchment on the receiving stream:

- *integrator site* in the impacted stream just above the confluence to measure the sum of the impacts in the catchment;
- *recovery site* downstream to measure the extent of recovery from the impact.

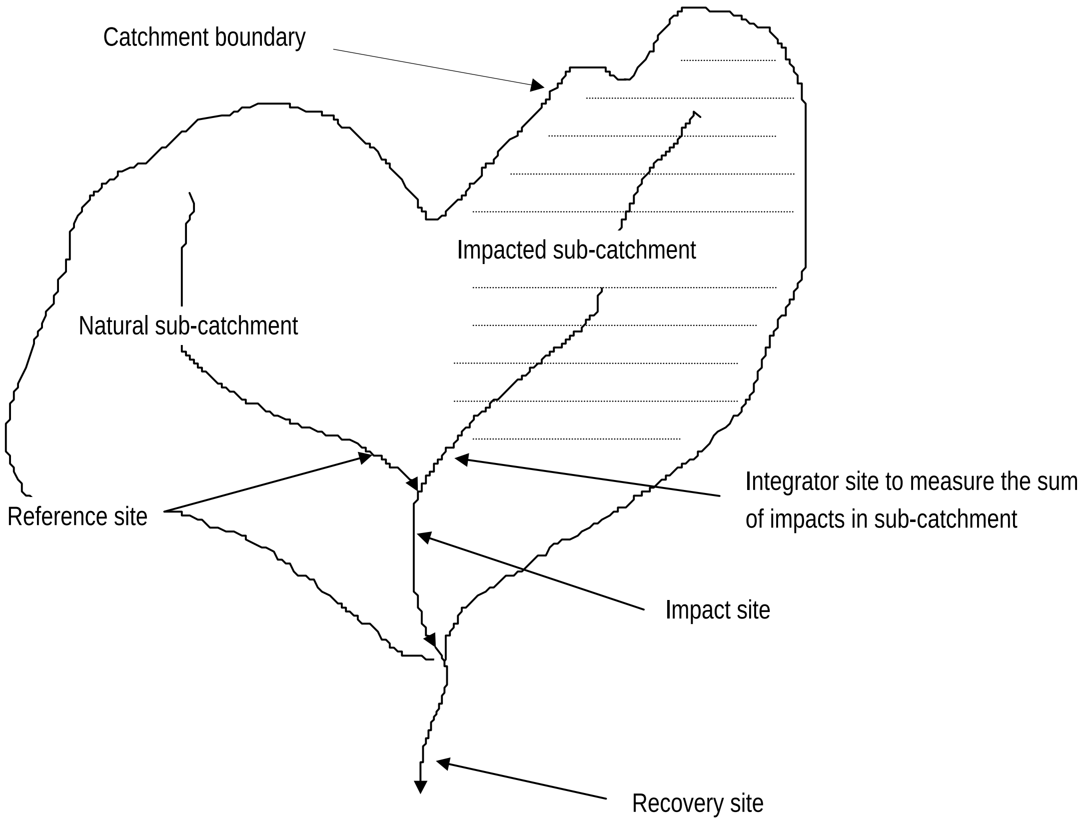


Figure 3. Location of sample sites for a paired catchment study

LAKES: SELECTING SITES

Although lakes lack the strong currents found in rivers, strikingly different conditions may exist throughout the lake depending on shape, embayments, depth, and prevailing winds.

General guidelines for selecting your sites

- The first step is to get a map of your lake showing depth contours, inlets and outlets. Use it to identify the sampling sites.
- The main sampling sites should be located at places that represent typical conditions of the lake. The best sites will be in the deepest part of the lake, or, in the case of a large lake, the deepest part of an arm or bay. Avoid near-shore areas, areas near inlets, secluded areas that may form lagoons, or areas downwind that may collect windblown algae and debris.
- Other sampling sites may be chosen to compare tributary inlets with outlets; extensively developed shorelines versus undeveloped areas; or seasonal camps with those of year-round occupation.
- Where possible, choose sites monitored by government agencies and sites in areas of public recreation, e.g. swimming.
- Layering (stratification) in the lake due to temperature differences will affect water quality readings. To check for stratification, you should sample at various depths.
- For estimates of the quantity of algae in the water, you should mix several samples together (composite sample) from the open water at the surface of the lake from sites 100 - 200m apart.

ESTUARIES: SELECTING SITES

Site selection in estuaries needs to account for the complex interactions between tides and river flows. Use a navigational chart that gives depth, eastings and northings, and navigational aids to select sites.

General guidelines for selecting your sites

- Where possible, choose sites monitored by government agencies and sites at areas of public recreation, e.g. swimming areas.
- Choose sites which are representative of the part of the estuary you are interested in.
- Select sites that are safely accessible.
- Because conditions in the main stream may differ from those at the edge, sites should be located in the main current and away from the banks or shoreline. If that is not possible, locate the site where thorough mixing of the water occurs, such as on an outside bend of the shoreline.
- Consider the possible effect of structures such as piers, groins and bulkheads on flow patterns. These may make your site unrepresentative of water quality in the estuary.
- Consider the influence of tides at your sampling location. You may need to sample at several different depths in order to get a representative sample, since fresh water from the river will "float" on the salt water coming in with the tide.

GROUNDWATER: SELECTING SITES

This depends on the type of aquifer system being investigated. If you are investigating an alluvial valley type aquifer system, it is prudent to sample a bore in each branch of the aquifer. It is also wise to take samples along the aquifer from the headwaters down to the point where the branch joins with main body of alluvium. Consider sampling bores outside the extent of alluvial aquifers to keep watch on the levels and quality of groundwater in the bedrock.

7.1 Record the details of site locations and mark each site on your catchment map.

7.2 List each sampling site and the reasons you used to select sampling sites.

8. When and how often will you monitor?

The purpose of monitoring is the key to knowing when and how often to monitor. Frequency of monitoring could be regular - weekly, monthly, seasonal, annual - or could be determined by irregular events, e.g. heavy rain. It depends on what you want to know and your resources. Your monitoring frequency should strike a balance between collecting too few data to detect important changes and collecting too much data so that trivial changes are detected.

CHOOSING THE BEST TIME AND FREQUENCY OF MONITORING

Baseline monitoring

Baseline monitoring is aimed at determining the baseline condition of the river and/or long term trends. Sampling or surveys should be done at regular intervals, e.g. monthly during base flow conditions at representative sites to characterise the waterway. If monitoring teams are deployed over the whole catchment at about the same time, you will produce a "snapshot" of conditions. Biological sampling should be conducted at the same time each year, because of seasonal variations in aquatic invertebrate populations and vegetation. Macro-invertebrate sampling is often done twice yearly but should be done no more than four times per year because extra sampling will deplete animal populations.

Monitoring to determine suitability of a waterway for a particular use

Monitor during times when the water is used, e.g. for irrigation or swimming. Your group may want to know if the water meets minimum standards for important uses: e.g. for swimming, test faecal bacteria during the warmer months.

Monitoring to determine pollution impacts

Frequent sampling is required when the aim is to determine if water quality standards are being exceeded. This reduces the chances of a spike in pollution levels going undetected.

Time your monitoring to coincide with the suspected impact. For example, if you want to know about the impact of sewage on aquatic ecosystem health, then you may choose to sample dissolved oxygen at sunrise when levels are at their lowest.

Water quality is often affected by rainfall. Monitoring during rain will catch pollutants as they are flushed from the land surface into the water. For example, if you are trying to establish a link between a suspected failing wastewater treatment plant and high bacteria levels, then sample during heavy rainfall.

To measure the load (mass transport) of sediment, or chemicals and nutrients carried by streams, you should sample during high flow events. Measurements taken during low flows may indicate only a small percentage of the total load transported by the stream.

In some catchments, all the sites may be affected by land use changes, and have no reference site available. In this case, you should compare data from before the change with data afterwards. However, there will always be some uncertainty as to whether the changes might have occurred naturally.

Monitoring groundwater

Because groundwater movement is relatively slow, chemical changes usually occur only slowly and sampling every one to six months may be adequate. Consider sampling before the winter rains and when they have finished. Where groundwater quality changes rapidly, it is best to sample more frequently. It is important to collect all your samples within a short period of time to provide a "snapshot" of groundwater quality. This can then be compared to previous snapshots.

QUESTIONS TO HELP YOU PLAN

- Will there be seasonal differences in natural conditions, land use or pollution discharge which affect your data?
- Will discharge levels (base flow, high flow) affect the parameter that you are monitoring?
- In rain event monitoring, how much rain is required before you go out to sample and how often will you sample during rain?
- In estuarine monitoring, will tidal cycles affect the timing of monitoring?
- Are releases of water from farm or hydro-electric dams planned for your waterway?
- How do macro-invertebrate life-cycles affect your monitoring schedule?
- Do you have to get samples to a lab before the holding time expires or the lab closes?
- Should sampling be done at certain times of day, e.g. sunrise?
- For school monitoring, how will teaching programs and limited duration of lessons affect site selection and monitoring?
- Will the changes you are looking for take longer to happen than the period of sampling?

WORKING OUT YOUR SCHEDULE

The main challenge on sampling days is to organise participants and keep track of each result.

Work out a schedule for your monitoring based on the following considerations:

- extra time on the first sampling day to cope with the inevitable last minute glitches!
- anticipated time for each test;
- maximum holding time for a given parameter between collection and testing the sample;
- back-up people for emergencies;
- travel time.

In short, be aware of the requirements for each of the procedures you have chosen in your monitoring plan. This will help you organise a realistic schedule for the sampling days and work out how each sample result will be tracked from the waterway to your database.

8 Describe the type of monitoring to be done; list the parameters, sampling frequency, period of sampling and time of day if appropriate.

9. Who will be involved and how?

Regional community networks can vary in size from tens to hundreds of volunteers scattered over the catchment. Consider questions such as who is going to collect and analyse samples, who will take part in surveying the habitat, who will coordinate activities, how will members be trained. You may find it helpful to refer back to **Involving others** in the *Part 1 Program Organising Guide*.

IDENTIFYING ROLES AND RESPONSIBILITIES

Consider all the tasks that should be done and encourage someone to fill each position. An easy way to display this is by using an organisational chart listing the positions and a brief description of what each position does. The position of data confidence officer is particularly important for groups aiming to produce high quality data.

9.1 List the titles of paid and unpaid roles and their responsibilities.

Planning project tasks

List the tasks to be done, the main person responsible and the due date.

9.2 List the tasks that describe what you are going to do.

Training members

Training in the consistent use of standard procedures is the best way to ensure that each participant is collecting credible data. Consult the *Part 7 Data Confidence Guidelines* for assistance on planning training sessions.

9.3 List answers to the questions - who, what, where and when.

How will sampling teams be organised?

Which groups will sample and where? Is there a group leader? How will they get to the sites? Is there a phone number to contact in case of emergency?

9.4 Describe how volunteers will be organised to collect data.

Looking after equipment

When you are ready to test the water it is very frustrating to find that the equipment is not performing: e.g. moisture has ruined reagents; the meter batteries are flat; the meter has not been calibrated, etc. Consider when and who will look after equipment.

Your equipment should be checked regularly and calibrated with fresh calibration solutions. See the *Part 7 Data Confidence Guidelines*. Calibration and inspection should be a part of group training.

9.5 List the inspection and calibration details for each piece of equipment to be used.

10. How will the data be managed and reported?

Data collected in the field by community group samplers is sent to the coordinator or data manager, screened and reported to the community and stake-holders as shown in **Figure 4. Data to information**.

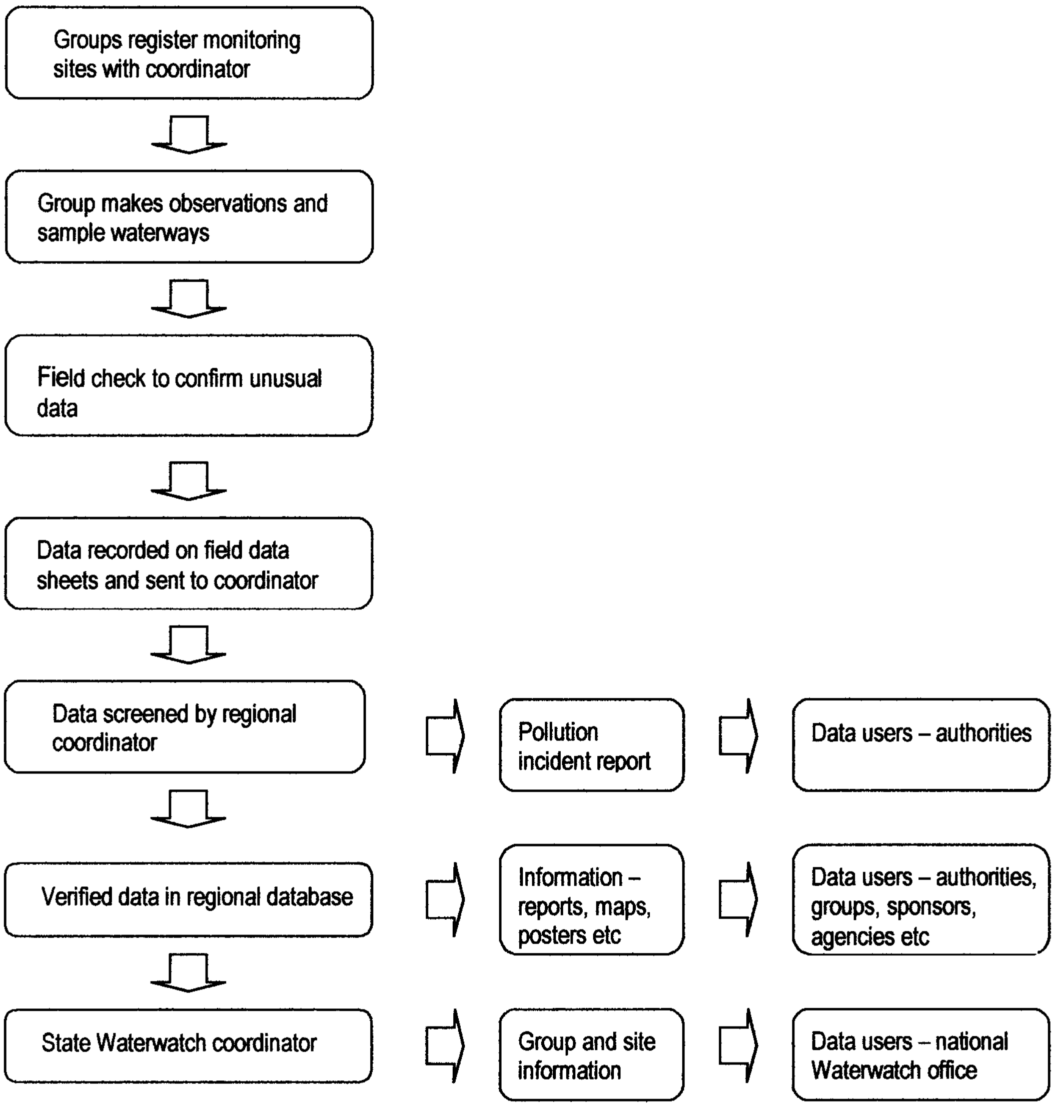


Figure 4. Data to information.

COLLECTING DATA ON SITE

When a water quality sample has been analysed, members should check that the result is within the expected range. If the result is outside the range, the test should be repeated to confirm the result. Confirmed results can then be entered on the field data sheet. A copy of the results is forwarded to the coordinator.

Use of standard record sheets helps to ensure consistency of data and comparability. During your training sessions, make sure you go over the field data sheets to be used. Many errors are made in transcribing data from instruments to field data sheets. Each reading recorded on the field data sheet must be verified by the sampler before being accepted as a true record.

10.1 Describe which field data sheets will be used by the group.

MANAGING REGULAR MONITORING DATA

Your group should have a clear plan for dealing with the data. Encourage a member to take on the role of data manager. Their role could include:

- keeping a complete and accurate record of all surveys and test results;
- making a copy and storing all records in a safe place;
- checking field data sheets for completeness and unusual results (outliers);
- entering checked data into the *Waterwatch* database program. See Question 11 - How will you ensure that your data are credible - for some pointers on checking your data.

Some monitoring groups may apply additional quality control checks over and above the minimum quality control checks employed by other groups (see *Part 7 Data Confidence Guidelines*). Keep the data from both groups separate when recording results. You will be able to use the data with additional checks with more confidence to interpret trends etc, and the other data to support your findings.

10.2 Describe how you store your data, photographs, original field data sheets.

MANAGING POLLUTION INCIDENT DATA

Guidelines that describe what to do if a pollution incident is discovered should be discussed and agreed to by the group before monitoring takes place. Pollution incident data should be reported to your coordinator who will immediately notify the local council and land manager or water authority. Your group may have an arrangement to notify downstream water users in the event of a pollution incident.

10.3 Describe the Pollution Incident Guidelines adopted by your group.

INTERPRETING THE DATA

Interpreting the data involves organising your data to show findings, and to develop conclusions and recommendations.

- **Findings** are observations about your data, e.g. which sites exceeded water quality standards and when.
- **Conclusions** are an explanation of why your data looks the way it does.
- **Recommendations** describe what action should be taken and what further information should be gathered.

COMMUNICATING THE DATA

Data doesn't mean a lot unless it is used, e.g. to educate, to make management decisions and as a basis for action. You should consider who will use the information and how before deciding on the best way of reporting your information. The users of your data could include the public, government officials and group participants.

Make a copy of the data file in your *Waterwatch* database program and regularly back it up.

See *Part 8 Moving From Data to Action* for more information on interpreting, reporting and acting on your results.

10.4 Describe who will use your information, how often you will report it, and what it will include.

11. How will you ensure that your information is credible?

Data users will accept or reject data based on the level of confidence they have in its reliability. To be credible, community group data has to be demonstrably reliable (fall within the designated tolerable error range) and be appropriate for its intended use.

Data confidence measures are the procedures that you use to ensure and assess the reliability of the information that you collect. They help you meet your data quality goals as described in **5. How accurate should the data be?** and ensure that users can have confidence in the data.

The following data confidence measures listed below are described in more detail in *Part 7 Data Confidence Guidelines*. They include:

- putting in writing all the steps taken to survey, monitor, analyse, store, manage and present data;
- encouraging trained individuals to be responsible for specific tasks that ensure that data quality is acceptable for its designated purpose;
- proper training, testing and retraining of participants;
- using quality control checks to continually assess the reliability of data and make necessary adjustments;
- reviewing your progress after an initial pilot stage;
- ensuring that the data are properly recorded on field data sheets and accurately transferred to the computer;
- following up unusual results.

QUALITY CONTROL CHECKS

The purpose of quality control (QC) checks is to let you know straight away if you have a problem so it can be corrected. They include both internal checks performed by the group members and coordinators and external checks performed by professional staff or independent labs. Data from external QC checks will carry more credibility than that from internal checks. Common types of quality control samples for water quality testing are described in the *Part 7 Data Confidence Guidelines*.

DATA CONFIDENCE LEVELS

Reliable data fall within the tolerable error range specified for the parameter. Reliable data justify a medium or high data confidence level depending on the type of QC checks employed. See **Figure 5. Data confidence levels**.

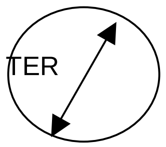
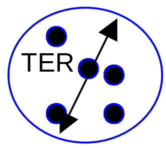
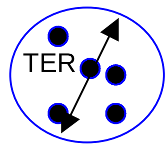
No data confidence	Medium data confidence	High data confidence
		
• No QC checks used. • No QC results available to enable comparison with tolerable error range (TER).	• Minimum internal QC checks used. • QC results fall within tolerable error range.	• Additional external QC checks used. • QC results fall within tolerable error range.

Figure 5. Data confidence levels. The size of the designated tolerable error range (TER) is determined by the monitoring goals of the group. The level of confidence in the data from the perspective of a data user who is not involved in collecting the results, is firstly determined by whether internal or external QC checks are used and secondly by how well the data satisfy those checks.

SUGGESTED DATA CONFIDENCE LEVELS AND QUALITY CONTROL CHECKS

To satisfy the diversity of needs of community groups, three broad levels of data confidence and associated QC checks are suggested below. The number and type of checks used will depend on the monitoring goal, resources and expertise within the group.

- **No QC checks – no data confidence**
  The absence of QC checks is appropriate for groups with a focus on education and raising awareness where the reliability of data is secondary to the actual process of collecting it. Since it is not known if data points are close to the true value, this data should not be added to the database. However, normal procedures should be used - samples are collected in the correct way, and equipment calibrated, kept clean and in good working order.

- **Minimum internal QC checks – medium data confidence level**
  Normal procedures and minimum quality control checks are used. Internal QC data points that fall within the designated tolerable error range indicate that the whole data set should be viewed with a medium level of confidence.

- **Additional QC checks – high data confidence level**
  In addition to normal procedures and quality checks listed above, 10% of samples are duplicates or replicates and tested by independent external agencies or labs. External QC data points that fall within the designated tolerable error range indicate that the whole data set should be viewed with a high level of confidence.

ENSURING RELIABLE DATA

Assessing reliability of data involves regularly comparing quality control results from mystery solutions, duplicate samples, etc. with the designated tolerable error range. If your data do not fall within the tolerable error range required, the problems can be fixed either by changing the method, improving training, or changing the acceptable tolerable error range. If you originally wanted to use the data for decision-making purposes, but it does not meet your requirements, it may still be useable for general or educational purposes.

11.1 Describe how you will screen your data and make decisions about accepting, rejecting or qualifying the data.

11.2 List the parameters that you will be sampling, and the type and number of quality control samples you will collect.

11.3 Describe how you will determine whether your data fall within the designated tolerable error range (see Question 5.2) and what you will do if data do not meet your requirements.

REVIEWING INITIAL PROGRESS.

Even with best-laid plans, things rarely go smoothly at first. Unanticipated problems, oversights and mistakes will all occur as the project proceeds. You should think of the early stages in the project as a "shake down" where the entire monitoring plan is tested in the field.

As problems arise and are solved, the monitoring plan should be revised. Revisions might include discarding some of your data; changing methods, equipment or field procedures; and providing more training.

Appendix 1
Extended monitoring plan record sheets

Congratulations! You have done a lot of work to get to this point. Now all that remains is to record your answers to the questions below.

The monitoring plan isn't finished when it is approved. It should influence every part of your program. Each person working on the project should be familiar with the plan, what is in it, and why it exists. Your answers will integrate data confidence measures and quality control with monitoring to produce a very important document that will guide every aspect of your community group activities.

Title page

Project Name ....
Community group name...
Affiliated organisations ....
Catchment(s) ....
Coordinator ....
Steering committee chairman ....
Other names (as needed) ....
Date ....

Contents page

Table of contents and page numbers.
Include a list of figures, tables, references and any appendices.

Page information

Each page should have information to keep track of changes. Monitoring plans are subject to frequent change and multiple editions and you need to keep track of which version is the most current.

Question numbers below correspond with explanatory notes in Section 2 of **Developing a monitoring plan.**

1. Why are you monitoring?

List the reasons for monitoring (objectives or specific question(s) you will try to answer).
You should include brief background information from your catchment survey that is relevant to the problem.
1.1 Reason for monitoring
1.2 Background information.

2. Who will use the data?

List the main users of your data.

3. How will the data be used?

Describe how the monitoring/survey data will be used by the data users.

4. What parameters will you monitor?

4.1 Describe your model using boxes and arrows.
4.2 Describe the parameters that you will survey or monitor. Describe how they will help you answer your question(s). Indicate if they are critical to the project or are collected for background information.

Parameter	How will it help to answer the question?	Critical / background

5. How accurate should the data be?

5.1 Describe the reporting units, resolution, detection limit and tolerable error range of the method used to test each parameter.

Parameter	Reporting units	Resolution	Detection limit	Tolerable error range

5.2. Describe how you will ensure that sampling is representative and comparable.

6. What methods will you use?

6.1 Complete the following table.

Parameter	Collecting procedure	Sample container	Preservation method (if applicable)	Maximum holding time	Equipment and method

7. Where will you monitor?

7.1 Record the details of site locations.

Site code	Map name	Map no.	Map scale	Easting (6 figs)	Northing (7 figs)	Catchment name	Sub-catchment name

7.2 List the sites to be surveyed/monitored and your reasons for selecting each site.

Site code	Site name	Site description	Reason for selecting site

8. When will you monitor?

List the type of monitoring to be done, parameters/ indicators, sampling frequency, period of sampling, and time of day (if appropriate).

Type of monitoring	Parameter	Sampling frequency	Period of sampling	Time of sampling

9. Who will be involved and how?

9.1 List the titles of paid and unpaid roles and their responsibilities.

Position	Responsibilities

9.2 List the tasks that you are going to do.

Project task	Main position responsible and supporting positions	Task completed by

9.3 In planning training, consider the questions - who, what, where and when.

Describe **who** will conduct group member training, who will attend training sessions, and who will evaluate the effectiveness of training.

- Trainer(s)
- Participants
- Evaluator(s)

Describe **what** the participants will learn and what they will be expected to demonstrate at the end of the training session(s). Describe **where** and **when** the training sessions will be held.

9.4 Describe how the volunteers will be organised to collect data. Is there a group leader? What is the emergency phone number to contact?

9.5 List the inspection and calibration details for each piece of equipment to be used in your program.

Equipment	Inspection frequency	Type of inspection	Calibration frequency	Standard or calibration instrument used

10. How will survey / monitoring data be managed and reported?

10.1 List which field data sheets your group(s) will be using.

10.2 Describe how you will store your data (photographs, original record sheets etc).

10.3 Outline the pollution incident guidelines adopted by your group.

10.4 Who will use your information, how often will you report it and what will it include?

11. How will you ensure that your data are credible?

11.1 Describe how you will screen your data and make decisions about accepting, rejecting or qualifying the data.

11.2 List the parameters that you will be sampling, and the type and number of quality control samples you will collect.

Parameter	Equipment and method	Quality control checks	QC sample frequency

11.3 Describe how you will determine whether your data fall within the designated tolerable error range (see Question 5.1) and what you will do if data do not meet your requirements.

Appendix 2
Example monitoring plan

1. Our reason for monitoring

1.1 Objective/question we will try to answer through monitoring.
Is stormwater runoff from the local township posing a risk to the health of canoeists who use the river, particularly during flood periods?

1.2 Relevant background information to our monitoring plan.
Rapids one kilometre downstream of the main storm water outlet for the town are a favourite spot for canoeists during floods. Upstream of the town, the river runs through bushland. Stormwater pipes discharge runoff from roads, parks and gardens directly into the river. Anecdotal evidence suggests that canoeists are becoming sick as a result of canoeing in the rapids. The local college is interested in collaborating with the canoe club to investigate water quality.

2. Main users of our data

The main users of our group's data will be the canoe club, secondary college and local council.

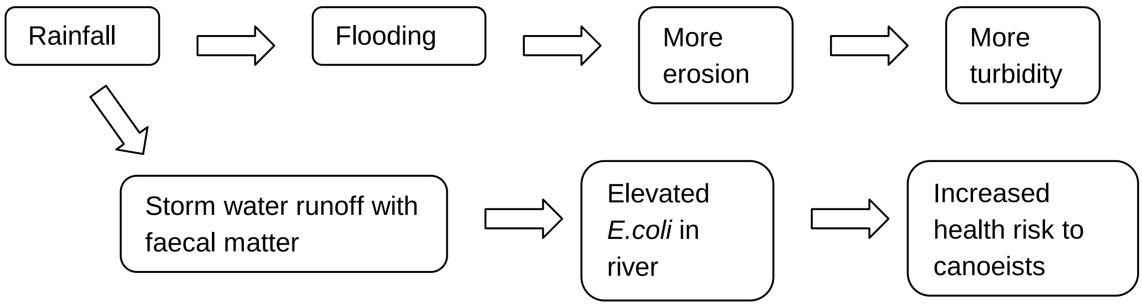
3. How the data will be used

Our group data will be used:

- *by canoe club members to gauge the risk to health during canoeing;*
- *to provide information to assist in council decisions that affect the river;*
- *to provide activities to enrich the learning experiences of college students.*

4 What we will monitor

4.1 Conceptual box and arrows model of the waterway problem.



4.2. Critical and background parameters that we will monitor and how they will help us answer the question.

Our group is interested in public health and has identified the following parameters:

Parameter	How will it help to answer the question?	Critical or background?
E.coli	*E.coli levels indicate the suitability of the river for canoeing.*	*Critical to answering the question.*
Turbidity	*Turbidity gives a general guide to water quality.*	*Background.*

5 Required data quality

5.1 Reporting units, resolution and detection limit and required tolerable error range of the method used to test each parameter.

Parameter	Reporting Units	Resolution	Detection limit	Tolerable error range
E.coli	*Colonies/100mLs*	*20 colonies/100mLs*	*20 colonies/100mLs*	*±200%**
Turbidity	*NTU*	*±10%*	*7 NTU*	*±20%*

** A wide tolerable error range is set for E.coli because of the inherently large variation in results from duplicate samples.*

5.2 How our sampling will give representative and comparable data.

- *Collect samples from upstream and downstream of the stormwater outlets and in the main rapids located downstream which are used by canoeists (representativeness)*
- *Samples will be collected and analysed using the same method (comparability).*

6 Methods to be used

6.1 Water quality sampling and analytical methods.

Parameter	Collecting procedure	Sample container	Preservation method	Maximum holding time	Equipment and method
E.coli	*Sample main current*	*Sterile whirl pack bag*	*Refrigerate*	*6 hours*	*Coliscan Easygel method*
Turbidity	*Sample main current.*	*Polyethylene 500ml*	*None required*	*24 hours*	*Turbidity tube*

7 Where we will monitor

7.1 Details of site locations.

Site code	Map name	Map no.	Map scale	Easting (6 figs)	Northing (7 figs)	Catchment name	Subcatchment name
SOM 010	*Somewhere*	*1234*	*1:100 000*	*460200*	*5414100*	*Somewhereelse River*	*not applicable*
SOM 020	*“*	*“*	*“*	*460900*	*5414700*	*“*	*“*

7.2 Sampling sites and the reasons for selecting those sites.

Site code	Site name	Site description	Reason for selecting site
SOM 010	*I'dratherbe somewhereelse*	*River 50m upstream of stormwater outlet*	*Reference site upstream of stormwater outlets. By comparing data from the reference site with that from the rapids, the source of bacteria will be identified.*
SOM 020	*Tipmeoutagain rapid*	*Grade 4 rapid starting at the large rock 1km below stormwater outfall*	*Site is at first main rapid used by canoeists downstream of stormwater outlets. This is where the greatest risk occurs to health of paddlers.*

8 Type of monitoring to be done; parameters, sampling frequency, period of sampling.

Type of monitoring	Parameter	Sampling frequency	Period of sampling	Time of sampling
Impact of storm water runoff	*E.coli*	*Irregular – rain event*	*During winter floods and canoeing*	*during heavy rain*
Nutrient runoff	*Turbidity*	*Irregular – rain event*	*During winter floods*	*during heavy rain*

9 Who is involved and how

9.1 Titles of paid and unpaid roles and their responsibilities.

Position	Responsibilities
Group coordinator	*Oversees the monitoring for the catchment; responsible for coordinating monitoring efforts for fifteen sub-groups in the catchment i.e. schools/Landcare groups; liaises with government agencies; obtains funding; coordinates work with communities; responsible for preparing monitoring plan and final reports; coordinates training of participants.*
Data confidence officer	*Oversees all data confidence and quality control activities; responsible for developing and updating the monitoring plan; supports the coordinator in training, checks quality of data and carries out follow up.*
Data manager	*Enter data onto the Waterwatch database from field sheets and prepares report for users.*
Participants: Grade 11 Environmental Science classes of the college.	*Responsible for implementation of the monitoring plan; carrying out surveys / sampling / analysis of data at the sites, forwarding data to data manager.*

9.2 Tasks, who is responsible and when to be completed.

Project task	Main position responsible and supporting positions	Task complete by:
Draft and get approval of monitoring plan	*Group coordinator, data confidence officer, agency quality assurance officer*	*February*
Ask potential members to become involved	*Group coordinator, data confidence officer, project coordinator*	*April*
Training sessions	*Group coordinator, data confidence officer*	*Three training sessions May*
Habitat survey	*College*	*May*
Water quality sampling	*College and canoe club*	*May*
Data processing and analysis	*Group coordinator, data confidence officer, data manager, group members*	*From June for 12 months to next June*
Reporting	*Group coordinator, data manager, group members*	*Media release within 1 week of each sampling event.* *Public meeting after 12 months of monitoring.*

9.3 Who, what, where and when.

The May training sessions will be run by the coordinator with help from the data confidence officer and for eight canoe club members and college teachers at the monitoring sites. All participants will be asked to evaluate the training sessions at the end of training.

All participants will be expected to demonstrate competence in safety, sampling technique, and testing for bacteria and turbidity.

9.4 How group members will be organised to collect data.

Students from Environmental Science at the College will be transported to sites SOM 010 and SOM 020 during rainfall events during term time. They will be under close supervision of the teacher when sampling. The emergency phone number is 12345 6789. They will be assisted by canoe club members.

9.5 Inspection and calibration details for each piece of equipment to be used.

Equipment type	Inspection frequency	Type of inspection	Calibration frequency	Standard or calibration instrument used
E.coli testing equipment	*Monthly*	*Media remain frozen until use.*	*Not applicable*	*Test a positive sample from each batch of 20 plates*
Turbidity tube	*Before each use*	*Visual check of cleanliness of tube.*	*Not applicable*	*Graduations determined from formazin standards.*

10 How the data will be managed and reported

10.1 Field data sheets that will be used by the group.

Schools in our group will use water quality data result sheets adapted from the Part 4 Water Quality Testing to record bacteria and turbidity.

10.2 How our data, photographs, original field data sheets will be stored.

Our data manager will enter data from field data sheets on the Waterwatch database. Record sheets will be stored and catalogued by site and date. A library of photos of all activities and sites in the catchment will be maintained.

10.3 Pollution Incident Guidelines adopted by our group.

Pollution incident guidelines for the group are:

1. *For unusual water quality results i.e. outside the normal range for the site, check calibration of the meter (if applicable), procedure and reagents used, and re-sample. If the original result is confirmed, record the result and immediately contact the coordinator so that water authorities and others in the catchment can be informed. For small regular incidents of this kind, keep a record of the time and day that it occurs for about a month and if possible try to identify the source.*
2. *If a severe pollution incident is discovered, contact the coordinator immediately (phone XXXX), who will contact the council. Severe incidents include:*
    - *large oil spills;*
    - *paint in the stream;*
    - *unusual colour change in the water;*
    - *fish kills, or kills of any animals;*
    - *large change in pH;*
    - *odours, an unusual appearance, fumes;*
    - *dumping of waste.*

10.4 Who will use our information, how often will we report it, and what it will include.

A workshop for members of the canoe club and the council will be held in June to present our year's data for discussion and action. We will release E.coli data after each sampling event to the general public to raise awareness of the problem.

11 How we will ensure credible information

11.1 How we will screen our data and make decisions about accepting, rejecting or qualifying the data.

Our data manager will continually review the completeness of data. They will screen data for transcription errors before entering onto the Waterwatch database. Decisions about accepting, rejecting or modifying data will be made in conjunction with the coordinator and data confidence officer.

11.2 Parameters that we will be sampling, and QC check samples to be collected.

Parameter	**Equipment and method**	**Quality control checks**	**QC sample frequency**
E.coli bacteria	*Coliscan Easygel*	*Positive plate to test viability of media*	*One plate per batch of 20 ordered from supplier*
		3 duplicate sample to test natural variation of FC numbers	*Prepare 3 duplicates from each site*
Turbidity	*Turbidity tube*	*Field replicate to measure natural variation and magnitude of error*	*Test a replicate at 10% of sites*

11.3 How we will determine whether the data fall within the designated tolerable error range (see Question 5.2) and what to do if the data do not meet our requirements.

The data confidence officer will check data quality as soon as possible after each sampling event. If data fall outside the tolerable error range, then reviews of training, methods, equipment and reagents will be done with the coordinator to identify the source(s) of the problem. The required standards of data quality may need to be revised. Any limitations on the use of the data will be reported to data users.

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